

STRENGTH OF FASTENED JOINTS IN COMPOSITES

Section: **3.6 – BOLTED JOINTS IN COMPOSITES**

Chapter: **3620**

Title: **STRENGTH OF FASTENED JOINTS IN COMPOSITES**

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1 INTRODUCTION

1.1 OBJECT, SCOPE AND LIMITATIONS

This chapter presents an overview of the methodology proposed to predict the strength of a bolted/fastened joint in plates made of fibrous composite laminates, when subjected to by-pass loads in plate and a load transfer from bolt to the plate.

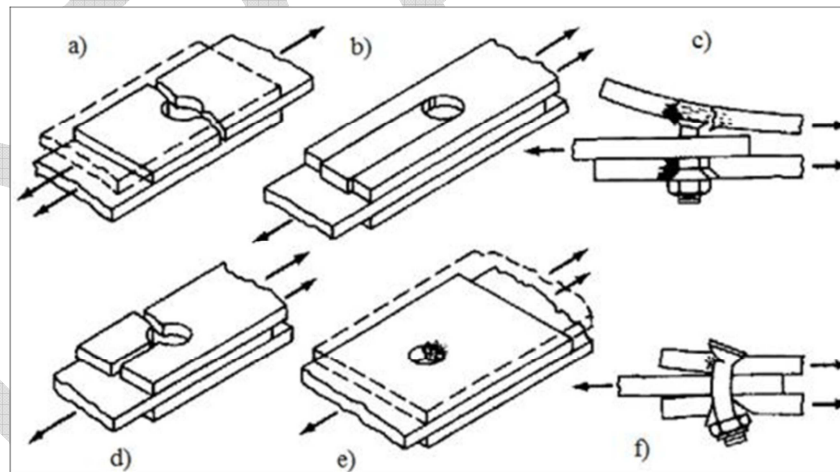


Figure 1. Typical failure modes in a fastened joint in composites

For considering the methodology applicable it is expected that the composite plates, and the fastened joint itself, follow the standard design and manufacturing rules (laminates stacking sequences, sufficient edge distance, drill tolerances, ...). Material shall be qualified and sufficient mechanical testing (bearing, filled-hole, ...) shall have been done to properly determine allowable design values.

The details of the methodology proposed and its implementation are described in Ref. 1. A brief overview is presented in following pages.

2 METHOD OVERVIEW

2.1 Loads in an fastened joint

Applied loads given, for a general case, as rivet shear/bearing load (Q at certain direction θ_{br}), by-pass/remote/far-field loads in terms of in-plane flows (N_x , N_y , N_{xy}), and tension force in the rivet (T)

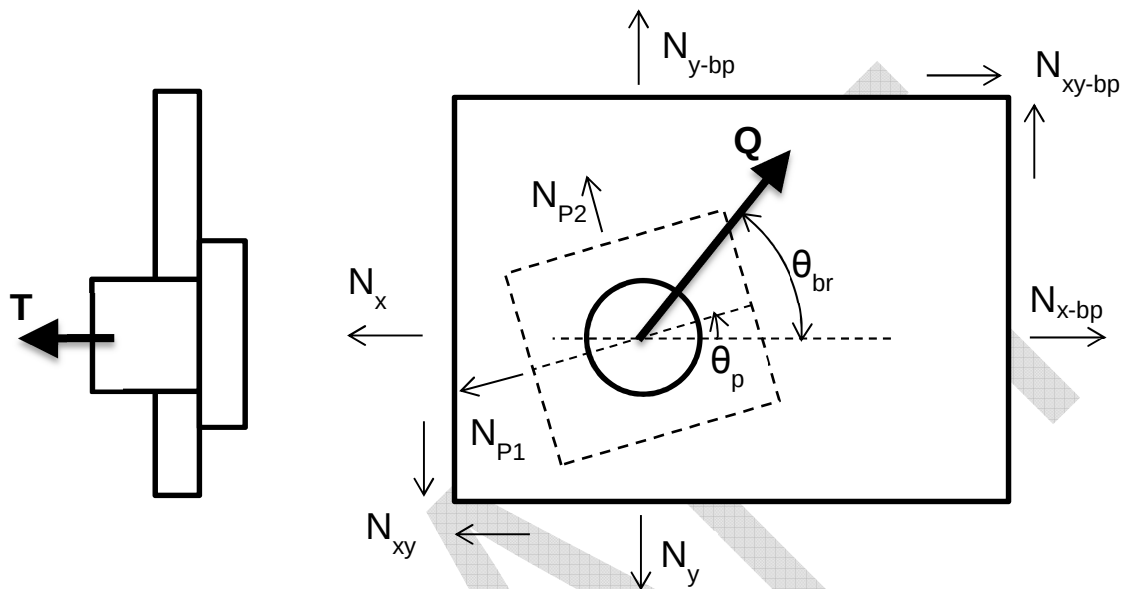


Figure 2. By-pass, shear/bearing and tension loads in a fastened joint

2.2 Net section failure

Stress distribution shall be evaluated at a discrete number of points around the hole (at the characteristic distance from hole edge), and the static margin calculated by means of the applicable plain strength criterion (point-stress approach). By-pass and tension-through-the-hole (bearing) cases shall be superimposed. Characteristic distance for the material is obtained using experimental data (FHT). Rest of tests experience and data (FHT, TTH, legacy methods) shall be used to correct calculated stresses or strengths.

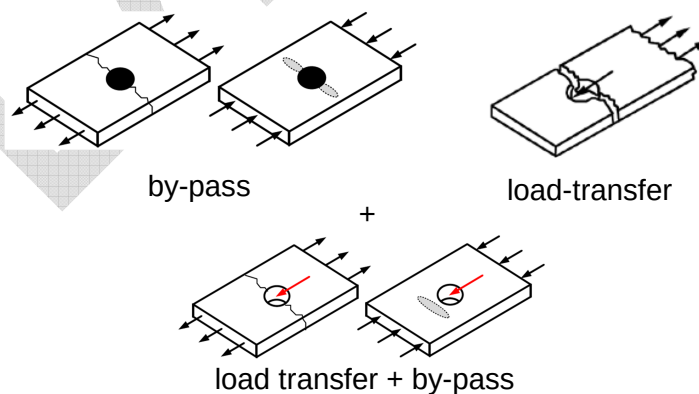


Figure 3. By-pass and Tension-Through-Hole failures

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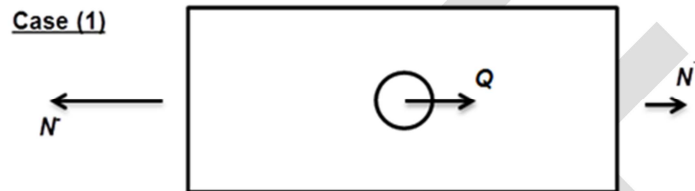
Stress distribution under by-pass loads

Analytical solutions of the stress/strain distributions in a composite (orthotropic) infinite plate with a circular hole under by-pass in-plane loads have been proposed by several authors, most based in the work of Lekhnitskii (see list of references).

The basic method proposed for computing these stresses in fastened joints (filled-hole) is the same as in the open hole configuration. Further discussion can be found in chapter §3404 "STRENGTH IN NOTCHED LAMINATES"

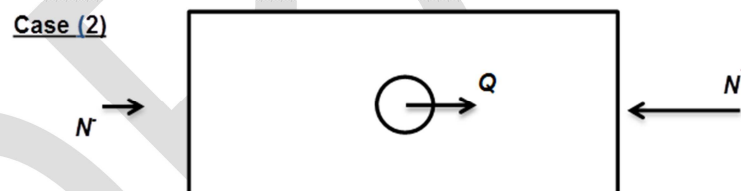
The determination of the by-pass loads in a general case, when the fastener is also transferring load, is not a simple case (see Figure 2). The following decision logic can be applied to longitudinal joints under axial loads (tension/compression lap-shear configurations):

- ✓ Case (1) : Tension load before and after the fastener



The load transfer at the pin Q is reacted at the side with the greater "far-field"/"gross" tension load (N in figure). The remainder must be equal to the tension at the side with lower load. Thus, the by-pass load is the lowest of the two tension values (N^+ in figure).

- ✓ Case (2) : Compression load before and after the fastener



Similar discussion as in Case (1). The by-pass load is the lowest (in terms of absolute values) of the two compression values (N in figure).

- ✓ Case (3) : Compression load at one side of the fastener and tension load at the other side



The loads system is only in equilibrium if the load transfer at the pin Q is capable to react the sum of the "far-field"/"gross" loads. Thus, the by-pass load is zero in this case.

Figure 4. Determination of by-pass loads in longitudinal joints

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For longitudinal joints under in-plane shear, the decision logic is identical: by-pass shear is zero if the signs of the in-plane shear are different before and after the fastener, and, if at both sides shear has the same sign, the by-pass shear is the lowest (in terms of absolute value) of the two sides.

Stress distribution under shear/bearing loads

For computing the stress distribution around a pinned loaded hole in a composite plate, the solution proposed by Garbo and Ogonowski (for the US Air Force Wright Aeronautical Laboratories), implemented in the code BJSFM (see Ref. 2), shall be adopted.

Further corrections

The influence of factors like the environmental conditions, finite edge distances (e/d , w/d), rivet head type (protruding or countersunk), type of joint (single or double shear configuration), eccentricity (t/d) in SLS unsupported joints, ... shall be taken into account as multipliers of the reference applied or allowable stresses, these multipliers expressed as independent monovariate functions of each of the factors (properly normalized to dimensionless variables). These functions are obtained as approximations of experimental test data and/or theoretical methods in literature.

2.3 Bearing failure

Bearing failure shall be referred to the bearing stress in the joint, as traditionally done (empirical approach), independently (within the in-plane analysis) of the stress concentration. The allowable value shall be referred to the appropriate load direction, considering the laminate stacking and rest of relevant correction factors.

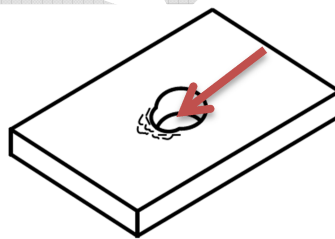


Figure 5. Bearing failure

Bearing stress is obtained from shear/bearing force, hole/pin diameter and plate thickness as:

$$\sigma_{br} = Q / (d \cdot t)$$

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2.4 Shear-out failure

Fastened joints following standard design rules, for this particular case with sufficient edge distance (i.e. $e/d > 2$), the shear-out failure shall not be a failure more liable to occur, and thus not required to be considered in strength check calculations.

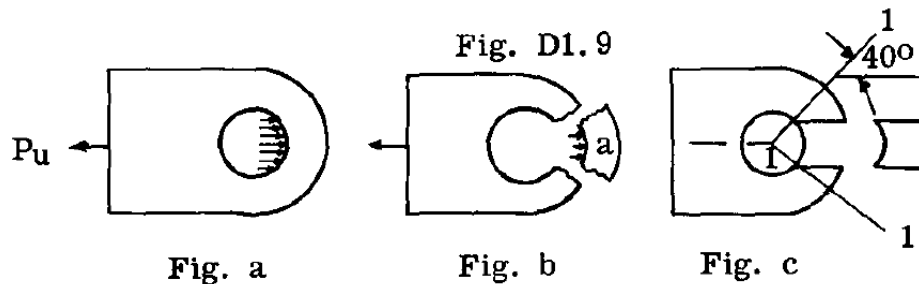


Figure 6. Shear-out in Bruhn's book, figure D1.9

In special configurations (e.g. composite lugs), Kassapoglou's proposal (see Ref. 10, eq. 11.38) shall be the recommended preliminary approximation (provided further test validation shall be required in any case for flyable parts).

2.5 Pull-out failure under bolt/rievet tension

Pull-out / Pull-Through failure under bolt tension shall be referred to the pull-out stress in the composite plate. The allowable value shall be available for each material and corrected appropriately for the type of rivet and the thickness to hole diameter ratio.



Figure 7. Pull-Out / Pull-Through failure

Pull-out stress is obtained from fastener tension force, bolt-head/nut external diameter and plate thickness as:

$$\tau_{po} = T / (\pi \cdot d_h \cdot t)$$

The tension force and the pull-out stress in this calculation shall have to account, in case the load is introduced by a tension-fitting alike configuration (as L-folded angles or T-fittings), for the following amplification effects:

- ✓ Tension force accounting for prying effects (out of the scope of this chapter, TBC)
- ✓ Pull-out stress concentrated on the side facing the "fitting" leg/web plate, typ. multiplying the pull-out stress in previous formula by a factor of $\times 2$

2.6 Interaction between Bearing and Pull-out

Being both matrix driven failures, it is expected that combined bearing and pull-out loads will produce failure below the failure loads for each of these failure modes considered uncoupled.

The combined margin shall be obtained by getting initially the uncoupled failure indices for bearing and pull-out (R in following formulas, the inverse of the reserve factors) and then applying the following interaction law:

$$R_{br}^{\alpha} + R_{po}^{\beta} = 1 \quad (\text{with } \alpha = \beta = 2, \text{ TBC})$$

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

Symbol	Units	Definition
BJSFM	-	Bolted Joints Stress Field Model (method and code proposed by Garbo and Ogonowski, see Ref. 2)
bp, by	-	As subscript, refers to by-pass load
BR, br	-	As subscript, refers to bearing
D, d	mm	Hole/pin diameter
dh	mm	Outer diameter of bolt head (or nut/collar)
DLS	-	Double Lap Shear
e	mm	Edge distance, i.e. distance from rivet centre to free edge (may be different for different directions)
FHC	-	Filled Hole Compression
FHT	-	Filled Hole Tension
N_x, N_y, N_{xy}	N/mm	In-plane by-pass forces per unit width
po	-	As subscript, refers to pull-out failure under bolt tension
Q	N	Shear/bearing force transferred by bolt/rivet
TBC	-	To Be Confirmed
t	mm	Thickness of the laminate
T	N	Tension force transferred by bolt/rivet
TTH	-	Tension Through the Hole
θ	°	Angle from X-plate / 0°-material direction to load/stress
ε	-	Strain
σ	MPa	Stress
τ	MPa	Shear stress
x, y	-	Directions of the reference axis system of the laminate
w	mm	Effective width of the plate, centred in fastener hole

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REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	TAE-XX-XX-XXXXXX	A	STRENGTH OF BOLTED JOINTS IN COMPOSITES
Ref. 2	AFWAL-TR-81-3041	1981	EFFECT OF VARIANCES AND MANUFACTURING TOLERANCES ON THE DESIGN STRENGTH AND LIFE OF MECHANICALLY FASTENED JOINTS. VOLUME 3 - BOLTED JOINT STRESS FIELD MODEL (BJSFM) COMPUTER PROGRAM USER's MANUAL
Ref. 3	[S. G. Lekhnitskii]	1968	Anisotropic Plates
Ref. 4	ESDU85001	1985	ELASTIC STRESS AND STRAIN DISTRIBUTIONS AROUND CIRCULAR HOLES IN INFINITE PLATES OF ORTHOTROPIC MATERIAL (Applicable to Fibre Reinforced Composites)
Ref. 5	[J. M. Whitney, R. J. Nuismer]	1974	Stress Fracture Criteria for Laminated Composites Containing Stress Concentrations
Ref. 6	[S.C. Tan]	1986	Laminated Composites Containing an Elliptical Opening. I. Approximate Stress Analyses and Fracture Models
Ref. 7	[S.C. Tan]	1986	Laminated Composites Containing an Elliptical Opening. II. Experiment and Model Modification
Ref. 8	[S.C. Tan]	1988	Finite-Width Correction Factors for Anisotropic Plate Containing a Central Opening
Ref. 9	[R.M. Jones]	2 nd Ed.	MECHANICS OF COMPOSITE MATERIALS
Ref. 10	[C. Kassapoglou]	2 nd Ed.	Design and Analysis of Composite Structures
Ref. 11	[E. F. Bruhn]	1973	ANALYSIS AND DESIGN OF FLIGHT VEHICLE STRUCTURES

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>

Table 1. Software programs referenced in this chapter

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RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.2 DD/05/2020	G. Baños	First issue	All pages

Table 2. Record of Revisions: authors, change reasons and sections/pages affected

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
<i>Stress Methods (TEASP-TL3)</i>	G. Baños DD/05/2020	F. Glock DD/MM/AAAA

Table 3. Approval signatures table for last revision of this chapter